Epilepsy Monitoring Project Update

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October, 2016

I. Hardware

Over the past few months, much of our focus has been determining unknown or implicit information from schematic files provided by device manufacturers. This knowledge is vital in order to optimize spatial constraints and reduce overall cost by eliminating unnecessary components. To achieve this, we have been collecting measurements of the component dimensions so we can ultimately aggregate them into a single PCB design using a production tool called EAGLE. Fortunately, RedBear Labs, the manufacturer of our Bluetooth module, has provided a `.dxf` file, shown in figure 1, which gives accurate information regarding size and placement [1]. This is essentially an image file that we can interact with using CAD software.

Recently, we have welcomed the help from a part-time student named Ken who works in PCB design at Boeing. He has been tremendously helpful in getting us closer to fabrication so that the components of our design can be mounted and electrically connected on a single part.

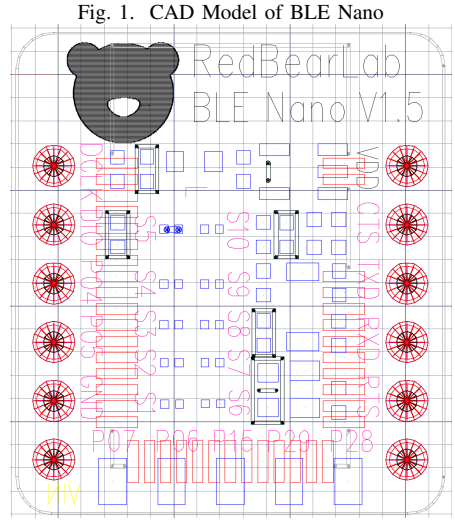
II. iOS

As a recent addition to the team, Jake has begun working on the user interface design for the mobile application. After completing several tutorials to learn Xcode, Apple's software development environment, Jake is looking to improve the existing codebase for the application. With the current application, we have established communication between the monitoring device using the Bluetooth Low Energy (BLE) protocol. The bare-bones user interface (UI) can display both heart rate and body temperature values in real time. As mobile development continues, intuitive and functional UI design will be a high priority. Ultimately, we hope to create a functional version of the app by the end of this semester.

III. Signal Processing

Due to the sensitive nature of monitoring one's vitals, reliable and efficient algorithms for noise reduction, artifact removal, and classification are paramount to the software design. Much of our work in this area has been in PPG signal analysis as a means of heart rate estimation. Previously, we considered the highly-effective TROIKA algorithm which can accurately estimate heart rate in the context of fitness trackers, where PPG signals are contaminated with strong motion artifacts [2]. Due to the novelty and relative complexity of the algorithm, we have postponed its implementation in favor of something with historical success in embedded devices [3], [4], [5].

Fig. 1. CAD Model of BLE Nano



For our preliminary PPG data processing, MATLAB was used to build a bandpass Butterworth filter, as suggested by the TROIKA algorithm. In essence, we are removing any high frequency noise and reducing the DC offset from the raw signal. Transients are removed after filtering and we then approximate the derivative using a first order backwards-difference, which is given by:

$$y[n] \approx x[n] - x[n-1]$$

MATLAB's peak detection algorithm is applied to the resulting signal to identify the number of peaks which lie above a threshold value. The heart rate (HR) is then calculated using:

$$\text{HR} = 60 \cdot \frac{\#\text{ of peaks}}{\text{elapsed time (sec)}}$$

It is worth noting that our HR calculations have given values two times larger than expected. This may be due to that delay in transmission as data are transferred over serial communication and written to a file, which was the process used for initial data collection. It is also possible that there's a small error in the firmware. In either case, this appears to be a systematic error and should be a simple fix.

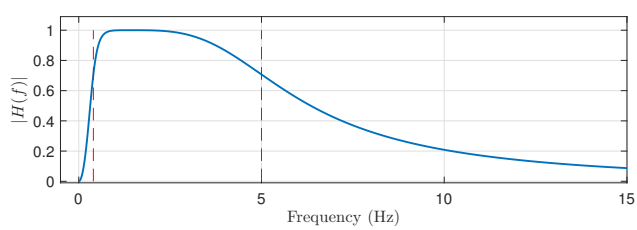
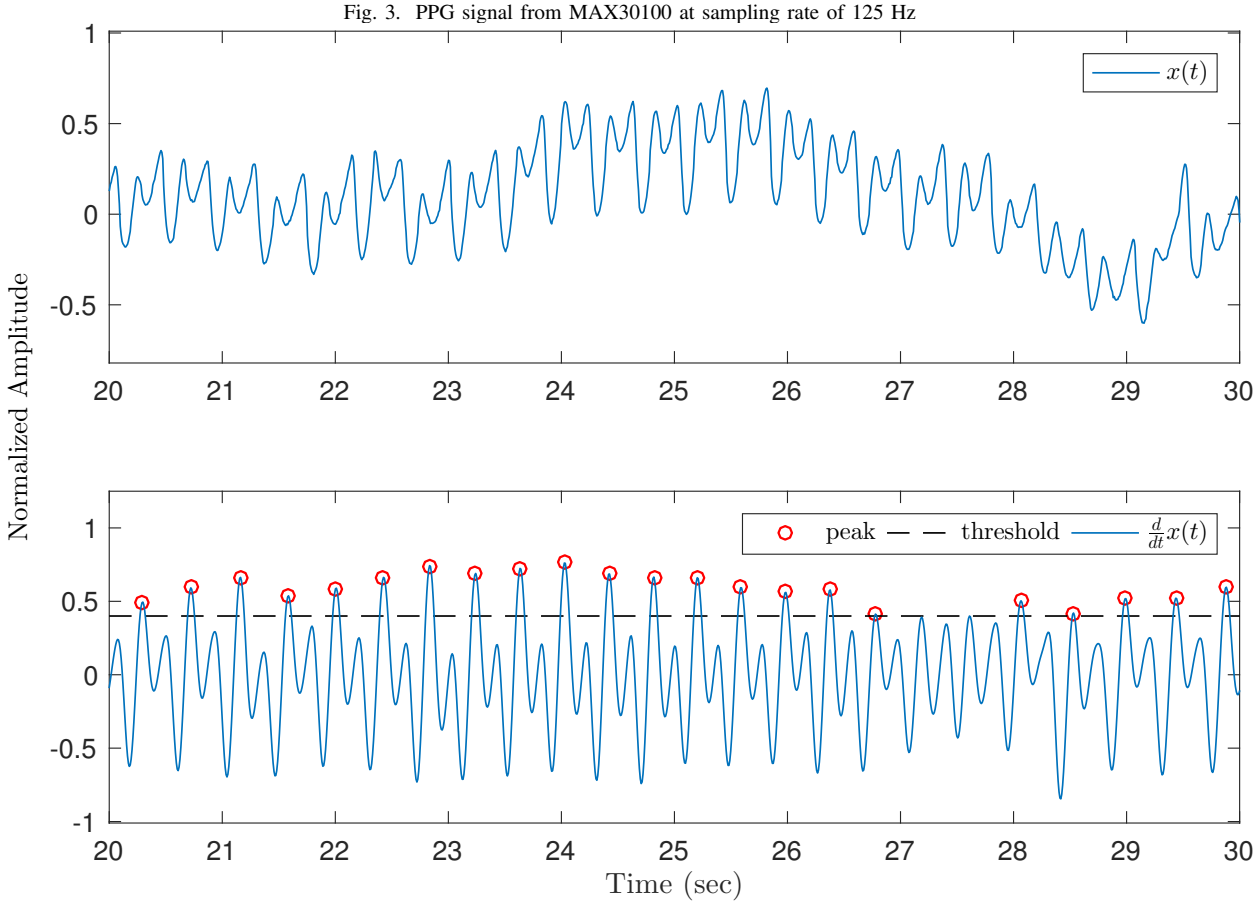


Fig. 2. Frequency Response of 2nd order Butterworth Filter with cutoff frequencies of 0.4 Hz. and 5 Hz.

Fig. 3. PPG signal from MAX30100 at sampling rate of 125 Hz



IV. Looking Ahead

In regards to heart rate estimation, we are looking into the autocorrelation function, defined by $R_{xx}(\tau)$. This will allow us to observe the relationship between energy and time delay as infrared reflectance data are sampled from the heart rate sensor. We would expect to see a dominant peak at the time delay which corresponds to the periodicity of the PPG signal. This is a fairly robust approach as seen in [6], and it eliminates the assumptions used for thresholding.

$$R_{xx}[\tau] = \sum_{k=-\infty}^{\infty} x[k]x[k+\tau]$$

One crucial aspect of this system will be its ability to reduce motion artifacts (MA). The plethysmographic (PPG) signals shown in figure 3 were recorded under stable conditions with little excess motion. We could improve the mechanically stabilize of the sensors, however this would most likely reduce comfort and wearability. A common technique used by researchers is to introduce an accelerometer in to the design. This gives information about the acceleration of the device in the X, Y, and Z directions. With this information, we can make inferences about the device's movement and make the appropriate processing decisions [3].

References

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[3] J. Guyomard and R. Stortelder, "Heart rate measurement through ppg: Heartbeat measurement in a wireless headset," 2015.

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